

Generation Interconnection Request System Impact Study Report

For Queue Project Q14-0358

Birchwood 138 kV
May 20, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0358. The Interconnection Customer proposes a 186.5 MW wind facility interconnecting at the Birchwood 138 kV substation in the Virginia area. The request was received on July 24, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 173 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,220,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Miller-Contra Costa 138 kV Cir. 2	Miller-Contra Costa 138 kV line	109.4	394	13.2
Fairland-Siuc 138 kV	Taber-Glenannan 345 kV	106.5	514	5.0
Miller-Contra Costa 138 kV	Contra Costa-Miller 138 kV Ckt 2	104.9	411	13.2
Taber 345/138 kV Transformer Bank	Dixleta-Taber Substation 345 kV	104.1	918	6.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Miller-Contra Costa 138 kV Cir. 2	Reconductor line	\$8,992,000	22% (\$1,980,000)
Fairland-Siuc 138 kV line	Reconductor line	\$690,000	3% (\$23,000)
Contra Costa-Miller 138 kV	Reconductor line	\$8,670,000	22% (\$1,909,000)
Taber 345/138 kV Transformer Bank	Replace transformer with higher-rated unit	\$17,972,000	2% (\$308,000)
	Total allocated to project		\$4,220,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.